

Firebase Authentication

1. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.MaterialComponents.Light">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="
                    android.intent.action.MAIN" />

                <category android:name="
                    android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name=".DashboardActivity"/>
        <activity android:name=".SignUpActivity"/>
    </application>

</manifest>
```

2. build.gradle (Project)

```
// Top-level build file where you can add configuration options common to all
sub-projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
    alias(libs.plugins.kotlin.android) apply false
    id("com.google.gms.google-services") version "4.4.3" apply false
}
```

3. build.gradle (module)

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
    id("com.google.gms.google-services")
}

android {
```

```

namespace = "polinema.mi.pml04"
compileSdk = 36

defaultConfig {
    applicationId = "polinema.mi.pml04"
    minSdk = 25
    targetSdk = 36
    versionCode = 1
    versionName = "1.0"

    testInstrumentationRunner =
        "androidx.test.runner.AndroidJUnitRunner"
}

buildTypes {
    release {
        isMinifyEnabled = false
        proguardFiles(
            getDefaultProguardFile("proguard-android-optimize.txt"),
            "proguard-rules.pro"
        )
    }
}

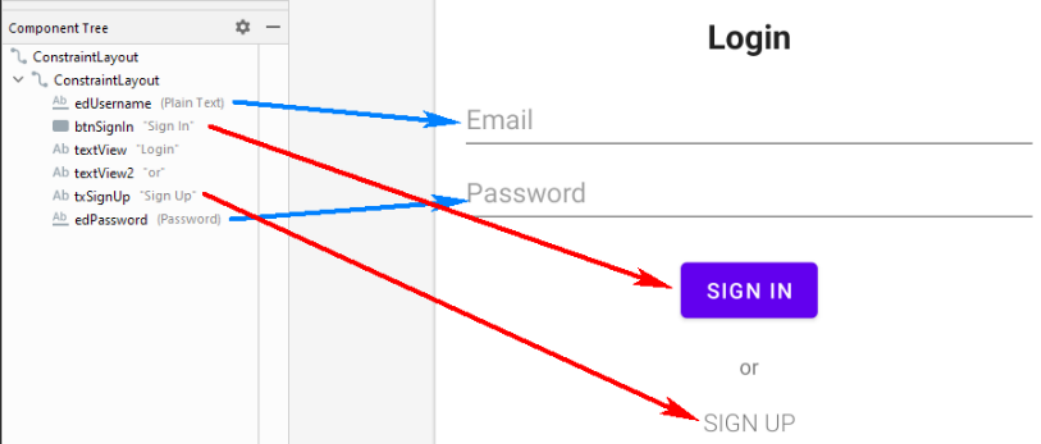
buildFeatures{
    viewBinding = true
}

compileOptions {
    sourceCompatibility = JavaVersion.VERSION_11
    targetCompatibility = JavaVersion.VERSION_11
}
kotlinOptions {
    jvmTarget = "11"
}

dependencies {
    implementation(platform("com.google.firebase:firebase-bom:34.3.0"))
    implementation("com.google.firebase:firebase-auth")
    implementation(libs.androidx.core.ktx)
    implementation(libs.androidx.appcompat)
    implementation(libs.material)
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}

```

4. activity_main.xml



```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <EditText
            android:id="@+id/edUsername"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginTop="16dp"
            android:layout_marginEnd="16dp"
            android:ems="10"
            android:hint="Email"
            android:inputType="textPersonName"
            android:minHeight="48dp"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@+id/textView" />

        <Button
            android:id="@+id/btnSignIn"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
```

```
        android:text="Sign In"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf=
            "@+id/edPassword" />
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Login"
    android:textAlignment="center"
    android:textAppearance=
        "@style/TextAppearance.AppCompat.Large"
    android:textStyle="bold"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="or"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf=
        "@+id/btnSignIn" />
```

```
<TextView
    android:id="@+id/txSignUp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Sign Up"
    android:textAllCaps="true"
    android:textAppearance=
        "@style/TextAppearance.AppCompat.Display4"
    android:textSize="16sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf=
        "@+id/textView2" />
```

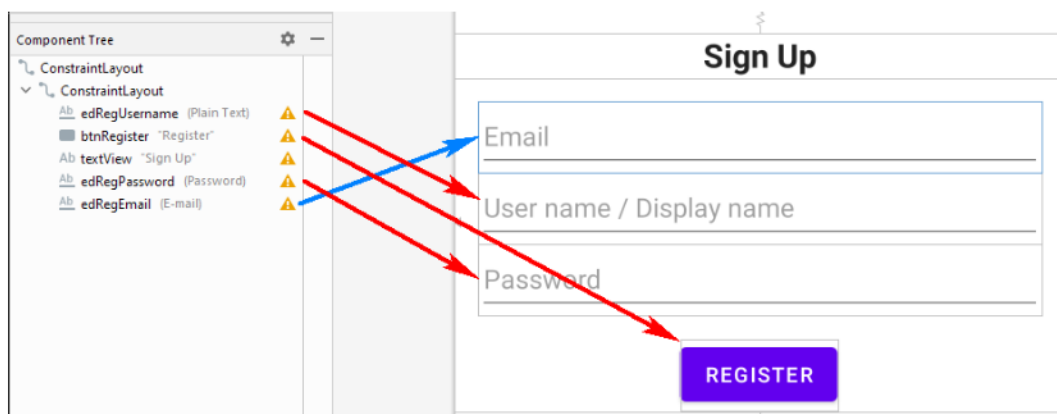
```
<EditText
    android:id="@+id/edPassword"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
```

```

        android:ems="10"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf=
            "@+id/edUsername" />
    </androidx.constraintlayout.widget.ConstraintLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

```

5. activity_signup.xml



```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <EditText
            android:id="@+id/edRegUsername"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginEnd="16dp"
            android:ems="10"
            android:hint="User name / Display name"

```

```
        android:inputType="textPersonName"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf=
            "@+id/edRegEmail" />
```

```
<Button
    android:id="@+id/btnRegister"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Register"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.498"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf=
        "@+id/edRegPassword" />
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Sign Up"
    android:textAlignment="center"
    android:textAppearance=
        "@style/TextAppearance.AppCompat.Large"
    android:textStyle="bold"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
<EditText
    android:id="@+id/edRegPassword"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:ems="10"
    android:hint="Password"
    android:inputType="textPassword"
    android:minHeight="48dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf=
        "@+id/edRegUsername" />
```

```
<EditText
    android:id="@+id/edRegEmail"
```

```

        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:ems="10"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:minHeight="48dp"
        app:layout_constraintEnd_toEndOf=
            "@+id/edRegUsername"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf=
            "@+id/edRegUsername"
        app:layout_constraintTop_toBottomOf="@+id/textView" />
    </androidx.constraintlayout.widget.ConstraintLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

```

6. activity_dashboard.xml

Component Tree

- ConstraintLayout
 - ConstraintLayout
 - btnProfil "Profil" ⚠
 - btnInputData "Input Data" ⚠
 - btnLogOut "Log Out" ⚠

PROFIL

INPUT DATA

LOG OUT

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <Button
            android:id="@+id/btnProfil"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:text="Profil"
            app:layout_constraintEnd_toEndOf="@+id/btnInputData"
            app:layout_constraintStart_toStartOf="@+id/btnInputData"
            app:layout_constraintTop_toTopOf="parent" />

        <Button
            android:id="@+id/btnInputData"

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Input Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/btnProfil" />

<Button
    android:id="@+id/btnLogOut"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Log Out"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="@+id/btnInputData"
    app:layout_constraintStart_toStartOf="@+id/btnInputData"
    app:layout_constraintTop_toBottomOf=
        "@+id/btnInputData" />
</androidx.constraintlayout.widget.ConstraintLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

```

7. MainActivity.kt (class)

```

package polinema.mi.pml04

import android.content.Intent
import android.os.Bundle
import android.view.View
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import com.google.firebase.Firebase
import com.google.firebase.auth.FirebaseAuth
import com.google.firebase.auth.FirebaseUser
import com.google.firebase.auth.auth
import polinema.mi.pml04.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity(), View.OnClickListener {

    lateinit var auth: FirebaseAuth
    lateinit var b : ActivityMainBinding
    var currentUser : FirebaseUser? = null

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivityMainBinding.inflate(layoutInflater)
        enableEdgeToEdge()
        setContentView(b.root)

        b.btnSignIn.setOnClickListener(this)
        b.txSignUp.setOnClickListener(this)

        auth = Firebase.auth
    }
}

```

```

    }

    override fun onClick(v: View?) {
        when(v?.id){
            R.id.btnSignIn ->{
                auth.signInWithEmailAndPassword(
                    b.edUsername.text.toString(),
                    b.edPassword.text.toString())
                .addOnCompleteListener {
                    if(it.isSuccessful){
                        currentUser = auth.currentUser
                        if(currentUser!=null){
                            if (currentUser!!.isEmailVerified){
                                Toast.makeText(this,
                                    "Login berhasil", Toast.LENGTH_LONG).show()
                                val intent =
                                    Intent(this,DashboardActivity::class.java)
                                startActivity(intent)
                            }else{
                                Toast.makeText(this,
                                    "Email anda belum terverifikasi",
                                    Toast.LENGTH_LONG).show()
                            }
                        }
                    }else{
                        Toast.makeText(this,
                            "Username/password salah",
                            Toast.LENGTH_LONG).show()
                    }
                }
            }
            R.id.txSignUp ->{
                val intent = Intent(this, SignUpActivity::class.java)
                startActivity(intent)
            }
        }
    }
}

```

8. DashboardActivity.kt (class)

```

package polinema.mi.pml04

import android.os.Bundle
import android.view.View
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import com.google.firebase.Firebase
import com.google.firebase.auth.FirebaseAuth
import com.google.firebase.auth.FirebaseUser
import com.google.firebase.auth.auth

```

```

import polinema.mi.pml04.databinding.ActivityDashboardBinding

class DashboardActivity : AppCompatActivity(), View.OnClickListener {
    lateinit var auth : FirebaseAuth
    lateinit var b : ActivityDashboardBinding
    var currentUser : FirebaseUser? = null

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivityDashboardBinding.inflate(layoutInflater)
        setContentView(b.root)

        b.btnProfil.setOnClickListener(this)
        b.btnInputData.setOnClickListener(this)
        b.btnLogOut.setOnClickListener(this)

        auth = Firebase.auth
    }

    override fun onClick(v: View?) {
        when(v?.id){
            R.id.btnProfil ->{
                currentUser = auth.currentUser
                if(currentUser!=null){
                    val builder = AlertDialog.Builder(this)
                    builder.setTitle("User Profile").setMessage(
                        "Nama : ${currentUser!!.displayName}\nEmail : ${currentUser!!.email}"
                    ).setNeutralButton("OK",null)
                    builder.show()
                }
            }
            R.id.btnLogOut ->{
                auth.signOut()
                finish()
            }
            R.id.btnInputData ->{
                //TODO : CALL SOME INTENT
            }
        }
    }
}

```

9. SignUpActivity.kt

```

package polinema.mi.pml04

import android.os.Bundle
import android.view.View
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.google.firebase.Firebase

```

```
import com.google.firebase.auth.FirebaseAuth
import com.google.firebase.auth.FirebaseUser
import com.google.firebase.auth.FirebaseAuth
import com.google.firebase.auth.userProfileChangeRequest
import polinema.mi.pml04.databinding.ActivitySignupBinding

class SignUpActivity : AppCompatActivity(), View.OnClickListener {
    lateinit var auth: FirebaseAuth
    lateinit var b: ActivitySignupBinding
    var currentUser: FirebaseUser? = null

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivitySignupBinding.inflate(layoutInflater)
        setContentView(b.root)

        b.btnRegister.setOnClickListener(this)
        auth = FirebaseAuth.getInstance()

    }

    override fun onClick(v: View?) {
        when(v?.id){
            R.id.btnRegister ->{
                auth.createUserWithEmailAndPassword(
                    b.edRegEmail.text.toString(),
                    b.edRegPassword.text.toString())
                    .addOnCompleteListener {
                        if(it.isSuccessful){
                            currentUser = auth.currentUser
                            if(currentUser!=null){
                                currentUser!!.updateProfile(
                                    userProfileChangeRequest {
                                        displayName = b.edRegUsername.text.toString()
                                    })
                                currentUser!!.sendEmailVerification()
                            }
                            Toast.makeText(this,
                                "Berhasil mendaftarkan user, silahkan cek email anda untuk memverifikasi akun anda",
                                Toast.LENGTH_LONG).show()
                            finish()
                        }else{
                            Toast.makeText(this, "Tidak dapat mendaftar",
                                Toast.LENGTH_SHORT).show()
                        }
                    }
            }
        }
    }
}
```

10. libs.version.toml (bila diperlukan)

```
[versions]
agp = "8.9.3"
kotlin = "2.0.21"
coreKtx = "1.17.0"
junit = "4.13.2"
junitVersion = "1.3.0"
espressoCore = "3.7.0"
appcompat = "1.7.1"
material = "1.13.0"
activity = "1.11.0"
constraintlayout = "2.2.1"

[libraries]
androidx-core-ktx = { group = "androidx.core", name = "core-ktx",
version.ref = "coreKtx" }
junit = { group = "junit", name = "junit", version.ref = "junit" }
androidx-junit = { group = "androidx.test.ext", name = "junit", version.ref =
"junitVersion" }
androidx-espresso-core = { group = "androidx.test.espresso", name =
"espresso-core", version.ref = "espressoCore" }
androidx-appcompat = { group = "androidx.appcompat", name =
"appcompat", version.ref = "appcompat" }
material = { group = "com.google.android.material", name = "material",
version.ref = "material" }
androidx-activity = { group = "androidx.activity", name = "activity",
version.ref = "activity" }
androidx-constraintlayout = { group = "androidx.constraintlayout", name =
"constraintlayout", version.ref = "constraintlayout" }

[plugins]
android-application = { id = "com.android.application", version.ref = "agp" }
kotlin-android = { id = "org.jetbrains.kotlin.android", version.ref = "kotlin" }
```

Referensi :

<https://firebase.google.com/docs/auth/android/email-link-auth>